ZOO203 - FINALTERM MERGED PPT
(LECTURE 36 TO 69)
Principles of Animal Life-II

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Social Behavior

Altruistic Behavior and Kin Selection

Altruistic Behavior and Kin Selection

- If, as Darwin suggested, animals should behave selfishly and strive to produce as many offspring as possible, why do some animals help others at some risk to themselves?
- Why do some individuals show utmost cooperation with members of their social group and even forego breeding themselves?
- Why do some individuals appear to sacrifice themselves so that other members of their group can survive?
- Until the mid 1960s, scientists had trouble explaining in Darwinian terms how such altruistic behavior could persist in a population.

Altruistic Behavior and Kin Selection

- Most instances of altruistic behaviors were explained using a groupselection argument.
- Group selectionists suggested that animals that helped others or that failed to mate did so for the benefit of the other members of the group.
- Therefore, such behaviors produce increased survivorship of groups whose members behaved altruistically.
- According to proponents of this argument, selection occurs at the level of the group, not at the level of the individual as Darwin suggested.

Altruistic Behavior and Kin Selection

- However, the group-selection argument as originally proposed by V. C. Wynne Edwards in 1962 has been rejected by the vast majority of behavioral ecologists for a number of reasons.
- For example, if in a social group there were randomly distributed genes for a risky altruistic behavior, such as giving calls to warn others of predators, those lacking such genes would flourish.
- They would be warned with no risk to themselves; their chances of reproduction would be greater and, in time, the “selfish” alleles would eliminate the altruistic ones from the group’s gene pool.

Altruistic Behavior and Kin Selection

- In 1964, W. D. Hamilton, based largely on his studies of insects, proposed a new way to explain altruistic behavior by modifying Darwin's original concept of fitness.
- He reasoned that fitness is measured not just by the number of offspring produced but by the increase or decrease in particular alleles in the gene pool.
- Thus, an individual may act altruistically, even at great risk, if it helps increase representation of its alleles in the gene pool.
- Alleles are shared by all relatives, including parents and offspring, brothers and sisters, cousins and other relations.

Altruistic Behavior and Kin Selection

- Alleles that influence altruistic behavior among relatives would persist in future generations.
- Since the most closely related animals share the most genes by common descent, we expect that altruistic behavior would be most common among closely related individuals.
- Thus, if everything else were equal, brothers who on average share half their alleles would be more likely to aid one another than they would a cousin who shares on average only 25% of their alleles.

Altruistic Behavior and Kin Selection

- Hamilton's hypothesis based on this genetic explanation for altruism and cooperation is called **kin selection**.
- Essentially, **kin selection** is the selection of genes by individuals assisting the survival and reproduction of individuals who possess the same genes by common descent.
- Hamilton's hypothesis revolutionized evolutionary and behavioral biology.
- The main criterion of Darwinian fitness is the relative number of an individual's alleles that are passed to future generations.

Altruistic Behavior and Kin Selection

- Hamilton, however, developed the concept of inclusive fitness, which is the relative number of an individual's alleles that are passed on to future generations either as a result of an individual's own reproductive success or that of related individuals.
- Thus, kin selection and inclusive fitness may be able to explain many altruistic behaviors that have perplexed biologists for many years.
- A good example of altruism and kin selection in nature is the remarkable cooperation and coordination among euscocial insects such as ants, bees, and wasps.

Altruistic Behavior and Kin Selection

- Through haplodiploidy, where males are haploid and females are diploid, sisters are genetically related on average by 75% rather than 50%.
- Sisters are more closely related to each other than to their own daughters!
- Therefore, they cooperate with other members of their social group, forego breeding themselves and aid the queen to produce more sisters who are more closely related (75% related) than potential offspring (50% related).
- Female Belding's ground squirrels, found in the High Sierra of California, give alarm calls when a predator approaches.
- Alarm calling warns other members of the social group and is risky to the alarm caller.

Altruistic Behavior and Kin Selection

- However, the benefits to alarm calling outweigh the risks because alarm callers are warning related individuals.
- Thus, alarm-calling behavior, even if it is risky, may be favored by selection if it increases inclusive fitness of the caller.
- Kinship theory suggests that animals may evolve an ability to recognize categories of relatives so that cooperation or aid-giving behavior will be directed more efficiently toward relatives.
- Although kin recognition behavior was discussed by Hamilton, little was known about it until almost 20 years after he wrote his seminal papers.

Altruistic Behavior and Kin Selection

- Through a number of experimental studies we now know that a variety of species can discriminate between kin and non-kin.
- Such species are found among invertebrates, including isopods and insects, and vertebrates, including fishes, frog and toad tadpoles, birds, squirrels, and monkeys.
- Some species can even discriminate between full siblings and half-siblings and between cousins and unrelated individuals.

Altruistic Behavior and Kin Selection

- Thus, some species have a finely tuned ability to identify relatives of various degrees of relatedness.
- The cues used in kin recognition vary from species to species.
- Birds use vocalizations whereas many other species use chemical cues.

Animal Communication

Animal Communication

- **Definition**
- **Animal communication** is the transfer of information from one or a group of **animals** (sender or senders) to one or more other **animals** (receiver or receivers) that affects the current or future behavior of the receivers.
- Only through communication can one animal influence the behavior of another.
- Compared with the enormous communicative potential of human speech, however, nonhuman communication is severely restricted.

Animal Communication

- **Means of Communication**
- Animals may communicate through following means:
- Sounds
- Scents
- Touch
- Movement

Animal Communication

- Indeed any sensory channel may be used, and in this sense animal communication has richness and variety.
- Unlike our language, which is composed of words with definite meanings that may be rearranged to generate an almost infinite array of new meanings and images, communication of other animals consists of a limited repertoire of signals.
- Typically, each signal conveys one and only one message.
- These messages cannot be divided or rearranged to construct new kinds of information.
- A single message from a sender may, however, contain several bits of relevant information for a receiver.

Animal Communication

- **Example:**
- The song of a cricket announces to an unfertilized female the species of the sender (males of different species have different songs), his sex (only males sing), his location (source of the song), and social status (only a male able to defend the area around his burrow sings from one location).
- This information is crucial to the female and accomplishes a biological function.
- But there is no way for the male to alter his song to provide additional information concerning food, predators, or habitat, which might improve his mate's chances of survival and thus enhance his own fitness

Animal Cmmunication

Chemical Sex Attraction in Moths

Chemical Sex Attraction in Moths

- **Mating**
- Mate attraction in silkworm moths illustrates an extreme case of stereotyped, single message communication that has evolved to serve a single biological function: mating.
- Virgin female silkworm moths have special glands that produce a **chemical sex attractant** to which males are sensitive.
- Adult males smell with their large bushy antennae, covered with thousands of sensory hairs that function as receptors as shown in figure below:

Chemical Sex Attraction in Moths



Chemical Sex Attraction in Moths

- **Bombykol**
- Most of these receptors are sensitive to the chemical attractant **bombykol** (a complex alcohol named after the silkworm Bombyx mori) and to nothing else.
- To attract males, females merely sit quietly and emit a minute amount of bombykol, which is carried downwind.
- When a few molecules reach a male's antennae, he is stimulated to fly upwind in search of the female.
- His search is at first random, but, when by chance he approaches within a few hundred yards of the female, he encounters a concentration gradient of the attractant.

Chemical Sex Attraction in Moths

- Guided by the gradient, he flies toward the female, finds her, and copulates with her.
- In this example of chemical communication the attractant bombykol, a pheromone, serves as a signal to bring the sexes together.
- Its effectiveness is ensured because natural selection favors the evolution of males with antennal receptors sensitive enough to detect the attractant at great distances (several miles).
- Males with a genotype that produces a less sensitive sensory system fail to locate a female and thus are reproductively eliminated from the population.

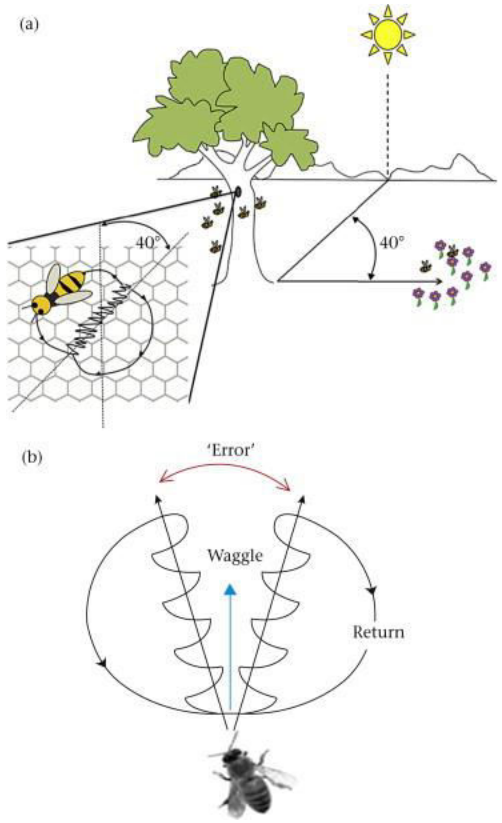
Animal Communication

Language of Honey Bees

Language of Honey Bees

- One of the most sophisticated and complex of all nonhuman communication systems is the symbolic language of bees.
- Honey bees are able to communicate the location of food resources when these sources are too distant to be located easily by individual bees.
- They communicate by dances, which are mainly of two forms.
- The form having the most informational richness is the waggle dance as shown in figure below.

Language of Honey Bees



Language of Honey Bees

- Bees most commonly execute these dances when a forager has returned from a rich source, carrying either nectar in her stomach or pollen grains packed in basketlike spaces formed by hairs on her legs.
- The waggle dance is roughly in the pattern of a figure-eight made against the vertical surface on the comb inside the hive.
- One cycle of the dance consists of three components:
- a circle with a diameter about three times the length of the bee
- a straight run while waggling the abdomen from side to side and emitting a pulsed, low frequency sound
- another circle, turning in the opposite direction from the first

Language of Honey Bees

- This dance is repeated many times with the circling alternating clockwise and counterclockwise.
- The straight, waggle run is the important informational component of the dance.
- Waggle dances are performed almost always in clear weather, and the direction of the straight run is related to the position of the sun as shown in the above figure.
- If the forager has located food directly toward the sun, she will make her waggle run straight upward over the vertical surface of the comb.

Language of Honey Bees

- If food was located 60 degrees to the right of the sun, her waggle run is 60 degrees to the right of vertical.
- We see then that the waggle run points at the same angle relative to the vertical as the food is located relative to the sun.
- Distance of the food source is also coded into bee dances.
- If the food is close to the hive (less than 50 m), the forager employs a simpler dance called the **round dance**.
- The forager simply turns a complete clockwise circle, then turns, and completes a counterclockwise circle, a performance that is repeated many times.

Language of Honey Bees

- Other workers cluster around the scout and become stimulated by the dance as well as by the odor of nectar and pollen grains from flowers she has visited.
- The recruits then fly out and search in all directions but do not stray far.
- The round dance carries the message that food is to be found in the vicinity of the hive.
- If the food source is farther away, round dances become waggle dances, which provide both distance and directional information.
- The tempo of the waggle dance is related inversely to the food's distance.

Language of Honey Bees

- If the food is about 100 m away, each figure-eight cycle lasts about 1.25 seconds; if 1000 m away, it lasts about 3 seconds; and if about 8 km (5 miles) away, it lasts 8 seconds.
- When food is plentiful, the bees may not dance at all.
- But when food is scarce, the dancing becomes intense, and other workers cluster around the returning scouts and follow them through the dance patterns.

Animals Communication

Communication by Displays

Communication by Displays

- **Definition**
- A display is a kind of behavior or series of behaviors that serves a communicative purpose.
- The release of sex attractant by the female moth and the dances of bees just described are examples of displays; so are alarm calls of herring gulls, songs of the whitecrowned sparrow, courtship dances of the sage grouse, and “eyespots” on the hind wings of certain moths that are exposed quickly to startle potential predators.

Communication by Displays

- The elaborate pair-bonding displays of blue-footed boobies as shown in figure below are performed with maximum intensity when the birds come together after a period of separation.
- The male at right in the illustration is sky pointing:
- the head and tail are pointed skyward and the wings are swiveled forward in a seemingly impossible position to display their glossy upper surfaces to the female.

Communication by Displays



Communication by Displays

- This display is accompanied by a highpiping whistle.
- The female at left, for her part, is parading. She steps with exaggerated slow deliberation, lifting each brilliant blue foot in turn, as though holding it aloft momentarily for the male to admire.
- Such highly personalized displays, performed with droll solemnity, appear comical, even inane to the observer.
- Indeed the boobies, whose name is derived from the Spanish word **“bobo”** meaning clown, presumably were so designated for their amusing antics.

Communication by Displays

- The exaggerated nature of the displays ensures that the message is not missed or misunderstood.
- Such displays are essential to establish and maintain a strong pair bond between male and female.
- This requirement also explains the repetitious nature of displays that follow one another throughout courtship and until laying of eggs.
- Redundancy of displays maintains a state of mutual stimulation between male and female, ensuring the degree

Communication between Humans and Other Animals

- One uncertainty in studies of animal communication is understanding what sensory channel an animal is using.
- The signals may be
- visual displays
- odors
- vocalizations
- tactile vibrations
- electrical currents (as, for example, among certain fishes).

Communication between Humans and Other Animals

- Even more difficult is establishing two-way communication between humans and other animals since the investigator must translate meanings into symbols that the animal can understand.
- Furthermore, people are poor social partners for most other animals.

PRE-DARWINIAN THEORIES OF CHANGE

PRE-DARWINIAN THEORIES OF CHANGE

Work of other scientists

- The idea of evolution did not originate with Charles Darwin.
- Some of the earliest references to evolutionary change are from the ancient Greeks.
- The philosophers **Empedocles** (495–435 B.C.) and **Aristotle** (384–322 B.C.) described concepts of change in living organisms over time.
- **Georges-Louis Buffon** (1707–1788) spent many years studying comparative anatomy.
- His observations of structural variations in particular organs of related animals convinced him that change must have occurred during the history of life on earth.

PRE-DARWINIAN THEORIES OF CHANGE

- Buffon attributed change in organisms to the action of the environment.
- He believed in a special creation of species and considered change as being degenerate—for example, he described apes as degenerate humans.
- **Erasmus Darwin** (1731–1802), a physician and the grandfather of Charles Darwin, was intensely interested in questions of origin and change.
- He believed in the common ancestry of all organisms.

Lamarck: an early proponent of evolution

- **Jean Baptiste Lamarck** (1744–1829) was a distinguished French zoologist.
- His contributions to zoology include important studies of animal classification.
- Lamarck published a set of invertebrate zoology books.
- His theory was based on a widely accepted theory of inheritance that organisms develop new organs, or modify existing organs, as needs arise. (Charles Darwin also accepted this idea of inheritance.)
- Similarly, he believed that disuse resulted in the degeneration of organs.

Lamarck: an early proponent of evolution

- Lamarck believed that “need” was dictated by environmental change and that change involved movement toward perfection. T
- The idea that change in a species is directed by need logically led Lamarck to the conclusion that species could not become extinct—they simply evolved into different species.
- Lamarck illustrated his ideas of change with the often quoted example of the giraffe.
- He contended that ancestral giraffes had short necks, much like those of any other mammal.

Lamarck: an early proponent of evolution

- Straining to reach higher branches during browsing resulted in their acquiring higher shoulders and longer necks.
- These modifications, produced in one generation, were passed on to the next generation.
- Lamarck published his theory in 1802 and included it in one of his invertebrate zoology books, Philosophie Zoologique (1809).
- He defended his ideas in spite of intense social criticism.

Lamarck: an early proponent of evolution

- Lamarck's acceptance of a theory of inheritance that we now know is not correct led him to erroneous conclusions about how evolution occurs.
- There is no evidence that changes in the environment can initiate changes in organisms that can be passed on to future generations.
- Instead, change originates in the process of gamete formation.
- Random changes in the structure of DNA (mutation) and chance processes involved in the assortment of genes into gametes (e.g., independent assortment, crossing-over, and random fertilization) result in variation among offspring.

Lamarck: an early proponent of evolution

- The environment then plays a role in determining the survival of these variations in subsequent generations.
- Even though Larmarck's mechanism of change was incorrect, he should be remembered for his steadfastness in promoting the idea of evolutionary change and for his numerous accomplishments in zoology.

DARWIN'S EARLY YEARS AND HIS JOURNEY

Voyage of the HMS Beagle

DARWIN'S EARLY YEARS AND HIS JOURNEY

- Charles Robert Darwin (1809–1882) was born on February 12, 1809.
- His father, like his grandfather, was a physician.
- During Darwin's youth in Shrewsbury, England, his interests centered around dogs, collecting, and hunting birds—all popular pastimes in wealthy families of nineteenth-century England.
- These activities captivated him far more than the traditional education he received at boarding school.
- At the age of 16 (1825), he entered medical school in Edinburgh, Scotland.

DARWIN'S EARLY YEARS AND HIS JOURNEY

- For two years, he enjoyed the company of the school's well established scientists.
- Darwin, however, was not interested in a career in medicine because he could not bear the sight of people experiencing pain.
- This prompted his father to suggest that he train for the clergy in the Church of England.
- With this in mind, Charles enrolled at Christ's College in Cambridge and graduated with honors in 1831.

DARWIN'S EARLY YEARS AND HIS JOURNEY

- This training, like the medical training he received, was disappointing for Darwin.
- Again, his most memorable experiences were those with Cambridge scientists.
- During his stay at Cambridge, Darwin developed a keen interest in collecting beetles and made valuable contributions to beetle taxonomy.

Voyage of the HMS Beagle

- One of his Cambridge mentors, a botanist by the name of John S. Henslow, nominated Darwin to serve as a naturalist on a mapping expedition that was to travel around the world.
- Darwin was commissioned as a naturalist on the HMS Beagle, which set sail on December 27, 1831, on a five-year voyage.
- Darwin helped with routine seafaring tasks and made numerous collections, which he sent to Cambridge.

Voyage of the HMS Beagle

- The voyage gave him ample opportunity to explore tropical rain forests, fossil beds, the volcanic peaks of South America, and the coral atolls of the South Pacific.
- Most important, Darwin spent five weeks on the Galápagos Islands, a group of volcanic islands 900 km off the coast of Ecuador.
- Some of his most revolutionary ideas came from his observations of plant and animal life on these islands.
- At the end of the voyage, Darwin was just 27 years old.

Voyage of the HMS Beagle

- By 1842, Darwin had developed the essence of his conclusions but delayed their publication because of uncertainty about how they would be received.
- His ideas were eventually presented before the Linnean Society in London in 1858, and On the Origin of Species by Means of Natural Selection was published in 1859 and revolutionized biology.
- In the years after his voyage, Darwin was an extremely prolific scientist.

Voyage of the HMS Beagle

- He published five volumes on Zoology of
- the Beagle Voyage (1843)
- Fertilization of Orchids (1862)
- The Variation of Plants and Animals under Domestication (1873)
- The Descent of Man (1871), and numerous other works

EARLY DEVELOPMENT OF DARWIN'S IDEAS OF EVOLUTION

Geology

EARLY DEVELOPMENT OF DARWIN'S IDEAS OF EVOLUTION

- The development of Darwin's theory of evolution by natural selection was a long, painstaking process.
- Darwin had to become convinced that change occurs over time.
- Before leaving on his voyage, Darwin accepted the prevailing opinion that the earth and its inhabitants had been created 6,000 years ago and had not changed since.
- Because his observations during his voyage suggested that change does occur, he realized that 6,000 years could not account for the diversity of modern species if they arose through gradual change.

EARLY DEVELOPMENT OF DARWIN'S IDEAS OF EVOLUTION

- Once ideas of change were established in Darwin's thinking, it took about 20 years of study to conceive, and thoroughly document, the mechanism by which change occurs.
- Darwin died without knowing the genetic principles that support his theory.

Geology

- During his voyage, Darwin read Charles Lyell's (1779–1875) Principles of Geology.
- In this book, Lyell developed the ideas of another geologist, James Hutton, into the theory of uniformitarianism.
- His theory was based on the idea that the forces of wind, rain, rivers, volcanoes, and geological uplift shape the earth today, just as they have in the past.
- Lyell and Hutton contended that it was these forces, not catastrophic events, that shaped the face of the earth over hundreds of millions of years.

Geology

- This book planted two important ideas in Darwin's mind:
- the earth could be much older than 6,000 years
- if the face of the earth changed gradually over long periods, could not living forms also change during that time?

Fossil Evidence

- Once the HMS Beagle reached South America, Darwin spent time digging in the dry riverbeds of the pampas (grassy plains) of Argentina.
- He found the fossil remains of an extinct hippopotamus-like animal, now called Toxodon, and fossils of a horselike animal, Thoatherium.
- Both of these fossils were from animals that were clearly different from any other animal living in the region.
- Modern horses were in South America, of course, but Spanish explorers had brought these horses to the Americas in the 1500s.
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Fossil Evidence

- The fossils suggested that horses had been present and had become extinct long before the 1500s.
- Darwin also found fossils of giant armadillos and giant sloths.
- Except for their large size, these fossils were very similar to forms Darwin found living in the region.

EARLY DEVELOPMENT OF DARWIN'S IDEAS OF EVOLUTION

Galápagos Islands

Galápagos Islands

- On its trip up the western shore of South America, the HMS Beagle stopped at the Galápagos Islands, which are named after the large tortoises that inhabit them (Sp. galápago, tortoise).
- The tortoises weigh up to 250 kg, have shells up to 1.8 m in diameter, and live for 200 to 250 years.
- The islands' governor pointed out to Darwin that the shapes of the tortoise shells from different parts of Albemarle Island differed.
- Darwin noticed other differences as well.

Galápagos Islands

- Tortoises from the drier regions had longer necks than tortoises from wetter habitats.
- In spite of their differences, the tortoises were quite similar to each other and to the tortoises on the mainland of South America.
- How could these overall similarities be explained?
- Darwin reasoned that the island forms were derived from a few ancestral animals that managed to travel from the mainland, across 900 km of ocean.
- Because the Galápagos Islands are volcanic and arose out of the seabed, no land connection with the mainland ever existed.

Galápagos Islands

- One modern hypothesis is that tortoises floated from the mainland on mats of vegetation that regularly break free from coastal riverbanks during storms.
- Without predators on the islands, tortoises gradually increased in number.
- Darwin also explained some of the differences that he saw.
- In dryer regions, where vegetation was sparse, tortoises with longer necks would be favored because they could reach higher to get food.

Galápagos Islands

- In moister regions, tortoises with longer necks would not necessarily be favored, and the shorter-necked tortoises could survive.
- Darwin made similar observations of a group of dark, sparrow-like birds.
- Darwin noticed that the Galápagos finches bore similarities suggestive of common ancestry.
- Scientists now think that Galápagos finches also descended from an ancestral species that originally inhabited the mainland of South America.

Galápagos Islands

- The chance arrival of a few finches, in either single or multiple colonization events, probably set up the first bird populations on the islands.
- Early finches encountered many different habitats containing few other birds and predators.
- Ancestral finches, probably seed eaters, multiplied rapidly and filled the seed-bearing habitats most attractive to them.
- Fourteen species of finches arose from this ancestral group, including one species found on small Cocos Island northeast of the Galápagos Islands.
- Each species is adapted to a specific habitat on the islands.

Galápagos Islands

- The most obvious difference between these finches relates to dietary adaptations and is reflected in the size and shape of their bills.
- The finches of the Galápagos Islands provide an example of adaptive radiation—the formation of new forms from an ancestral species, usually in response to the opening of new habitats.
- Darwin's experiences in South America and the Galápagos Islands convinced him that animals change over time.

Galápagos Islands

- It took the remaining years of his life for Darwin to formulate and document his ideas, and to publish a description of the mechanism of evolutionary change.

THE THEORY OF EVOLUTION BY NATURAL SELECTION

THE THEORY OF EVOLUTION BY NATURAL SELECTION

- On his return to England in 1836, Darwin worked diligently on the notes and specimens he had collected and made new observations.
- He was familiar with the obvious success of breeders in developing desired variations in plant and animal stocks.
- He wondered if this artificial selection of traits could have a parallel in the natural world.
- Ideas of how change occurred began to develop on his voyage.

THE THEORY OF EVOLUTION BY NATURAL SELECTION

- They took on their final form after 1838 when he read an essay by Thomas Malthus (1766–1834) entitled Essay on the Principle of Population.
- Malthus believed that the human population has the potential to increase geometrically.
- (Geometric growth involves increasing by doubling or by some other multiple rather than by adding a fixed number of individuals with each new generation.)
- However, because resources cannot keep pace with the increased demands of a burgeoning population, population-restraining factors, such as poverty, wars, plagues, and famine, begin to have an influence.

THE THEORY OF EVOLUTION BY NATURAL SELECTION

- Darwin realized that a similar struggle to survive occurs in nature.
- When viewed over generations, this struggle could be a means of natural selection.
- Traits that were detrimental for an animal would be eliminated by the failure of the animal containing them to reproduce.

THE THEORY OF EVOLUTION BY NATURAL SELECTION

Natural Selection

Natural Selection

- Charles Darwin had no knowledge of modern genetic concepts, and therefore had no knowledge of the genetic principles that are the basis of evolutionary theory as it exists today.
- The modern version of his theory can be summarized as follows:
- All organisms have a far greater reproductive potential than is ever realized.
- For example, a female oyster releases 100,000 eggs with each spawning, a female sea star releases about 1 million eggs each season, and a female robin may lay four fertile eggs each season.

Natural Selection

- What if all of these eggs were fertilized and developed to reproductive adults by the following year?
- A half million female sea stars (half of the million eggs would produce females and half would produce males), each producing another million eggs, repeated over just a few generations, would soon fill up the oceans!
- Even the adult female robins, each producing four more robins, would result in unimaginable resource problems in just a few years.

Natural Selection

Inherited variations

- **Inherited variations** exist.
- They arise from a variety of sources, including mutation, genetic recombination and random fertilization.
- Seldom are any two individuals exactly alike.
- Some of these genetic variations may confer an advantage to the individual possessing them.
- In other instances, variations may be harmful to an individual.

Natural Selection

- In still other instances, particular variations may be neither helpful nor harmful. (These are said to be neutral.)
- These variations can be passed on to offspring.
- Because resources are limited, existence is a constant struggle.
- Many more offspring are produced than resources can support; therefore, many individuals die.
- Darwin reasoned that the individuals that die are those with the traits (variations) that make survival and successful reproduction less likely.

Natural Selection

Adaptive traits

- Traits that promote successful reproduction are said to be adaptive.
- Adaptive traits become more common in subsequent generations.
- Because organisms with maladaptive traits are less likely to reproduce, the maladaptive traits become less frequent in a population.
- With these ideas, Darwin formulated a theory that explained how the tortoises and finches of the Galápagos Islands changed over time.
- In addition, Darwin's theory explained how some animals, such as the ancient South American horses, could become extinct.

Natural Selection

- What if a group of animals is faced with a new environment to which it is illadapted?
- **Reasons of extinction:**
- Climatic changes
- food shortages
- other environmental stressors
- All these could lead to extinction.

THE THEORY OF EVOLUTION BY NATURAL SELECTION

Adaptation

Adaptation

- Adaptation occurs when a heritable change in a phenotype increases an animal's chances of successful reproduction in a specified environment.
- Adaptations must be heritable changes to be passed to subsequent generations.
- Adaptations are defined in the context of enhancing reproductive success because survival of a species occurs through successful reproduction, and survival of the species is the ultimate measure of success.
- Adaptations are defined in the context of a specified environment because a change that promotes successful reproduction in one environment may be detrimental to reproductive success in a different environment.

Adaptation

- Even though adaptations are defined in the context of reproductive success, they can manifest themselves in a variety of ways.
- Adaptations may be behavioral, physiological, or morphological.
- For example, arctic animals display many adaptations to their environment.
- The arctic hare (Lepus arcticus) acquires a snow-white coat in the winter as camouflage from predators.
- Its relatively small ears (as compared to other hares) help prevent heat loss as blood circulates through peripheral vessels of its ears.

Adaptation

- Its huge feet (also in comparison to those of other hares) help keep the hare on top of the snow as it moves around the arctic tundra.
- All of these adaptations promote survival and make successful reproduction more likely.
- New adaptations arise as a result of mutations, and they are perpetuated by natural selection.
- Mutations are chance events, and most mutations are either harmful or neutral (see chapter 3).
- Adaptive mutations never occur as a result of a need, and there is no guarantee that a species will change in order to meet the challenges of a changing environment.

Adaptation

- If adaptive changes did occur in response to need, extinction would not occur—and extinction is a fact of life for the majority of species.
- Most genetic variations exist as neutral alleles, having arisen by mutation years earlier, and are expressed as adaptive traits only when a population encounters a new environment and natural selection acts on the population.
- Adaptation may result in the evolution of multiple new groups if the environment can be exploited in different ways.
- When the evolution of multiple groups occurs, adaptive radiation results.

Adaptation

- Not every characteristic is an adaptation to some kind of environmental situation.
- An allele that provided some adaptive trait in one environment may be neutral when the environment changes, but persist in the population because the trait is not detrimental.
- Other alleles may result in traits that were never adaptive.
- Because these alleles result in neutral traits, they are not selected against by natural selection and persist in the population.

MICROEVOLUTION, MACROEVOLUTION, AND EVIDENCE OF MACROEVOLUTIONARY CHANGE

MICROEVOLUTION, MACROEVOLUTION, AND EVIDENCE OF MACROEVOLUTIONARY CHANGE

- Organic evolution was defined earlier as a change in populations over time, or simply “descent with modification.”
- The change must involve the genetic makeup of the population in order to be passed to future generations.
- These observations have led biologists to look for the mechanisms by which changes occur.
- There is no doubt that genetic changes in populations occur—they have been directly observed in the field and in the laboratory.

MICROEVOLUTION, MACROEVOLUTION, AND EVIDENCE OF MACROEVOLUTIONARY CHANGE

- These changes are the reason that bacteria gain resistance to antibiotics and agricultural pests become resistant to pesticides.

Microevolution

- A change in the frequency of alleles in populations over time is called microevolution.
- Over longer timescales, microevolutionary processes result in large-scale changes.

Macroevolution

- Large-scale changes that result in extinction and the formation of new species are called macroevolution.

MICROEVOLUTION, MACROEVOLUTION, AND EVIDENCE OF MACROEVOLUTIONARY CHANGE

- **Macroevolutionary** changes are difficult to observe in progress because of the geological timescales that are usually involved.
- Evidence that macroevolution occurs, however, is compelling.
- This evidence is in the form of patterns of plant and animal distribution, fossils, biochemical molecules, anatomical structures, and developmental processes.
- Just like a burglar cannot carry out a crime without leaving some kind of evidence behind, organisms leave evidence of what they looked like and how they lived.

MICROEVOLUTION, MACROEVOLUTION, AND EVIDENCE OF MACROEVOLUTIONARY CHANGE

- Evolutionary investigators piece this evidence together and provide detailed accounts of the lives of extinct organisms and their relationships to modern forms.
- The sources of evidence for macroevolution are described in the next section.

MICROEVOLUTION, MACROEVOLUTION, AND EVIDENCE OF MACROEVOLUTIONARY CHANGE

Biogeography

Biogeography

DEFINITION

- Biogeography is the study of the geographic distribution of plants and animals.
- Biogeographers try to explain why organisms are distributed as they are.
- Biogeographic studies show that life-forms in different parts of the world have distinctive evolutionary histories.

Biogeography

- One of the distribution patterns that biogeographers try to explain is how similar groups of organisms have dispersed to places separated by seemingly impenetrable barriers.
- For example, native cats are inhabitants of most continents of the earth, yet they cannot cross expanses of open oceans.
- Obvious similarities suggest a common ancestry, but similarly obvious differences result from millions of years of independent evolution.

Biogeography

- Bio geographers also try to explain why plants and animals, separated by geographical barriers, are often very different in spite of similar environments.
- For example, why are so many of the animals that inhabit Australia and Tasmania so very different from animals in any other part of the world?
- The major native herbivores of Australia and Tasmania are the many species of kangaroos (Macropus).
- In other parts of the world, members of the deer and cattle groups fill these roles.

Biogeography

- Similarly, the Tasmanian wolf/tiger (Thylacinus cynocephalus), now believed to be extinct, was a predatory marsupial that was unlike any other large predator.
- Finally, biogeographers try to explain why oceanic islands often have relatively few, but unique, resident species.
- They try to document island colonization and subsequent evolutionary events, which may be very different from the evolutionary events in ancestral, mainland groups.
- The discussion that follows will illustrate some of Charles Darwin's conclusions about the island biogeography of the Galápagos Islands.

Biogeography

- Modern evolutionary biologists recognize the importance of geological events, such as volcanic activity, the movement of great landmasses, climatic changes, and geological uplift, in creating or removing barriers to the movements of plants and animals.
- Biogeographers divide the world into six major biogeographic regions.
- As they observe the characteristic plants and animals in each of these regions and learn about the earth's geologic history, we understand more about animal distribution patterns and factors that played important roles in animal evolution.
- Only in understanding how the surface of the earth came to its present form can we understand its inhabitants.

MICROEVOLUTION, MACROEVOLUTION, AND EVIDENCE OF MACROEVOLUTIONARY CHANGE

Paleontology

Paleontology

Definition

- **Paleontology** (Gr. palaios, old 1 on, existing 1 logos, to study), which is the study of the fossil record, provides some of the most direct evidence for evolution.

Fossils

- Fossils (L. fossilis, to dig) are evidence of plants and animals that existed in the past and have become incorporated into the earth's crust (e.g., as rock or mineral).
- Fossils are formed in sedimentary rock by a variety of methods.

Paleontology

- Most commonly, fossilization occurs when sediments (e.g., silt, sand, or volcanic ash) quickly cover an organism to prevent scavenging and in a way that seals out oxygen and slows decomposition.
- As sediments continue to be piled on top of the dead organism, pressures build.
- Water infiltrates the remains and inorganic compounds and ions replace organic components.
- Hard parts of the organism are most likely to be fossilized, but delicate structures are sometimes fossilized when silica is involved with replacement.

Paleontology

- These pressure and chemical changes transform the organism into a stony replica.
- Other fossilization processes form molds, casts of organisms, or carbon skeletons.
- Tracks and burrows, and even mummified remains, are sometimes found.
- Fossilization is most likely to occur in aquatic or semiaquatic environments.
- The fossil record is therefore more complete for those groups of organisms living in or around water and for organisms with hard parts.

Paleontology

- The fossil record provides information regarding sequences in the appearance and disappearance of organisms.
- Different strata of rock result from differing rates of sedimentation.
- Climatic and geological events influence rates of sedimentation. When rates of sedimentation change, a break in deposition occurs, leaving a distinct layer or stratum.
- Successive strata are piled on top of each other, with younger strata on top of older strata.

Paleontology

- Fossils in younger strata are of animals that lived more recently than fossils in older strata.
- Geologists can correlate strata around the world and determine when strata were formed using radioactive dating.
- Paleontologists use this information to provide an understanding of many evolutionary lineages.
- Many vertebrate lineages are very well documented in the fossil record.
- For example, the fossil record allows the history of horses to be traced back about 55 million years.

Paleontology

- Most of the evolutionary events occurred in what is now North America.
- Hyracotherium was a dog-sized animal (0.2 m in height at the shoulder).
- Fossils reveal the presence of four prominent toes on each foot and a tooth structure indicative of a browsing lifestyle.
- As the habitat became more grassland-like, natural selection favored animals with longer legs used for outrunning predators and larger more durable teeth (molars) used for grazing.
- A loss of some foot bones, and a reduction in others, was accompanied by an elongation of the middle digit.

Paleontology

- The shift from browsing to grazing was also accompanied by an elongation of the face.
- Paleontologists have also used the fossil record to describe the history of life on earth.
- Evidence from paleontology is clearly some of the most convincing evidence of macroevolution.

MICROEVOLUTION, MACROEVOLUTION, AND EVIDENCE OF MACROEVOLUTIONARY CHANGE

Analogy and Homology

Analogy and Homology

- Structures and processes of organisms may be alike.
- There are two reasons for similarities, and both cases provide evidence of evolution.
- Resemblance may occur when two unrelated organisms adapt to similar conditions.
- For example, adaptations for flight have produced flat, gliding surfaces in the wings of birds and insects.

Analogy and Homology

- These similarities indicate that independent evolution in these two groups produced superficially similar structures and has allowed these two groups of animals to exploit a common aerial environment.

Convergent Evolution

- The evolution of superficially similar structures in unrelated organisms is called convergent evolution, and the similar structures are said to be **analogous** (Gr. analog 1 os, proportionate).
- Resemblance may also occur because two organisms share a common ancestry.

Analogy and Homology

Homologous

- Structures and processes in two kinds of organisms that are derived from common ancestry are said to be homologous (Gr. homolog 1 os, agreeing) (i.e., having the same or similar relation).
- Homology can involve aspects of an organism's structure, and these homologies are studied in the discipline called **comparative anatomy**.
- Homology can also involve aspects of animal development and function, and homologous processes are studied using techniques of molecular biology.

MICROEVOLUTION, MACROEVOLUTION, AND EVIDENCE OF MACROEVOLUTIONARY CHANGE

Comparative Anatomy

Comparative Anatomy

Definition

- **Comparative anatomy** is the study of the structure of living and fossilized animals and the homologies that indicate evolutionarily close relationships.
- In many cases, homologies are readily apparent.
- For example, vertebrate appendages have a common arrangement of bones that can be traced back through primitive amphibians and certain groups of fish.
- Appendages have been modified for different functions such as swimming, running, flying, and grasping, but the basic sets of bones and the relationships of bones to each other have been retained.

Comparative Anatomy

- The similarities in structure of these bones reflect their common ancestry and the fact that vertebrate appendages, although modified in their details of structure, have retained their primary functions of locomotion.
- Other structures may be homologous even though they differ in appearance and function.
- The origin of the middle ear bones of terrestrial vertebrates provides an example.

Comparative Anatomy

- Fish do not have middle and outer ears.
- Their inner ear provides for the senses of equilibrium, balance, and hearing, with sound waves being transmitted through bones of the skull.
- Terrestrial vertebrates evolved from primitive fish, and the evolution of life on land resulted in an ear that could detect airborne vibrations.
- These vibrations are transmitted to receptors of the inner ear through a middle ear and, in some cases, an outer ear.

Comparative Anatomy

- One (amphibians, reptiles, and birds) or more (mammals) small bones of the middle ear transmit vibrations from the eardrum (tympanic membrane) to the inner ear.
- Studies of the fossil record reveal the origin of these middle ear bones.
- Small bones involved in jaw suspension in primitive fish are incorporated into the remnants of a pharyngeal (gill) slit to form the middle ear.
- In amphibians, reptiles, and birds, a single bone of fish (the hyomandibular bone) forms the middle ear bone (the columella or stapes).

Comparative Anatomy

- In the evolution of mammals, two additional bones that contributed to jaw support in ancient fish (the quadrate and articular bones) are used in the middle ear (the incus and malleus, respectively).
- There are many other examples of structures that have changed from an ancestral form. Sometimes these changes result in a structure that can be detrimental to the organism.
- The human vermiform appendix evolved from a large fermentation pouch, and it is still used in this manner in animals like rabbits and many other herbivores.
- In humans, the vermiform appendix may have functions related to the lymphatic system, but these functions pale in comparison to the problems that result when it becomes infected.

Comparative Anatomy

- Boa constrictors have minute remnants of hindlimb (pelvic) bones that are remnants of the appendages of their reptilian ancestors.
- Baleen whales, like all whales, evolved from land mammals and also retain remnants of the pelvic appendages their ancestors used for walking on land.
- In both of these cases the remnant pelvic appendages serve no apparent function.

Vestigial Structures

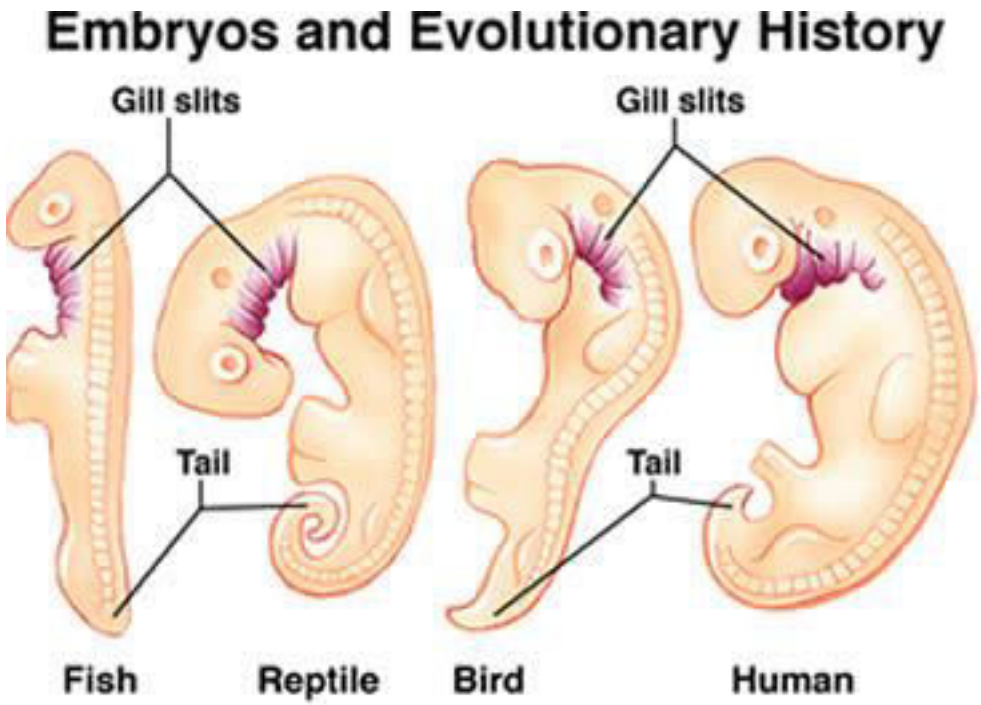
- Structures that have no apparent function in modern animals but clearly evolved from functioning structures in ancestors are called vestigial structures and provide another source of evidence of evolutionary change.

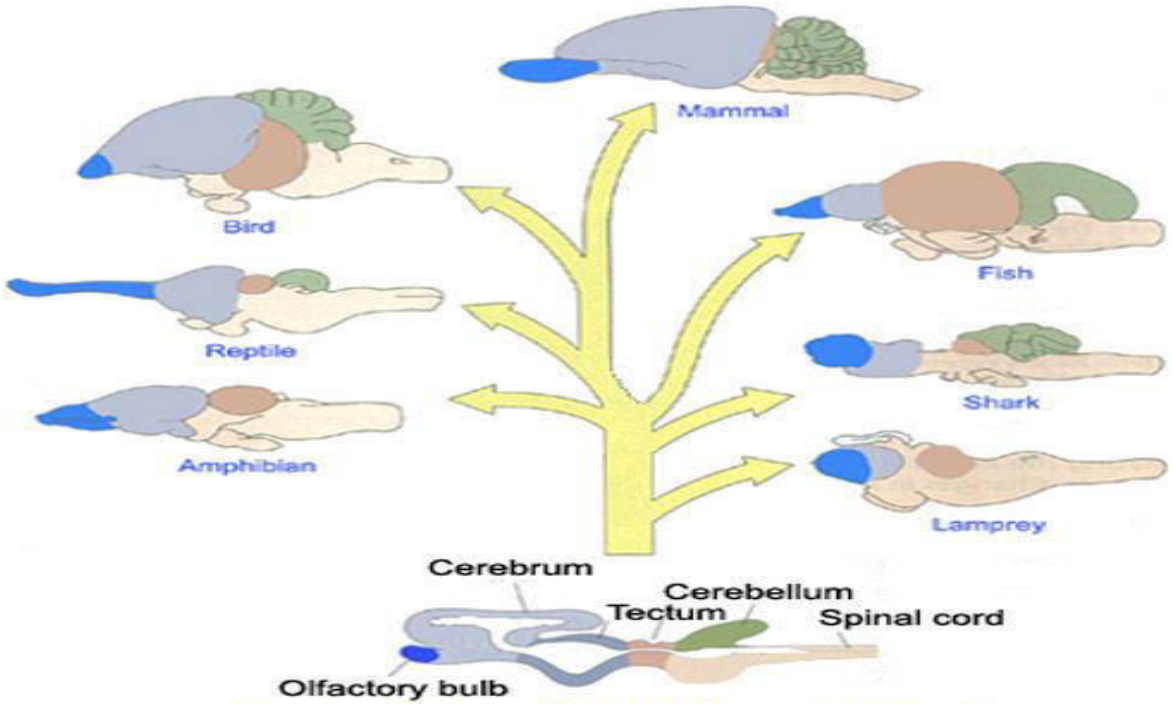
MICROEVOLUTION, MACROEVOLUTION, AND EVIDENCE OF MACROEVOLUTIONARY CHANGE

Developmental Patterns

Developmental Patterns

- Evidence of evolution also comes from observing the developmental patterns of organisms.
- The developmental stages of related animals often retain common features because changes in the genes that control the development of animals are usually harmful and are eliminated by natural selection.
- For example, early embryonic stages of vertebrates are remarkably similar as shown in figure.
- Many organ systems of vertebrates also show similar developmental patterns as shown in figure below.


Embryos and Evolutionary History
Gill slits
Gill slits
Tail
Tail
Fish
Reptile
Bird
Human



Basic Plan of Vertebrate Brain

Developmental Patterns

- These similarities are compelling evidence of evolutionary relationships within animal groups.
- Adults in vertebrate groups are obviously different from one another.
- If developmental patterns are so similar, how did differences in adult stages arise?
- Evolution is again the answer.
- These differences are a result of evolutionary changes in the genes that control the onset of developmental stages and the rate at which development occurs.

Developmental Patterns

- These changes result in differences in the size and proportions of organs.
- Differing growth rates of the bones of the skull, for example, can explain the proportional differences between bones of the human and ape skulls.
- Modern developmental biology is providing a growing appreciation of how evolution has conserved many genes that control the developmental similarities of animal groups.
- At the same time, it is helping to explain developmental changes that result in the great diversity of animal life.

MICROEVOLUTION, MACROEVOLUTION, AND EVIDENCE OF MACROEVOLUTIONARY CHANGE

Molecular Biology

Molecular Biology

- Within the last 25 years, molecular biology has yielded a wealth of information on evolutionary relationships.
- Studying changes in anatomical structures and physiological processes reflects genetic change and evolution.
- Unfortunately, it is often difficult to sort out the relationship between genes and the structures and functions they control.
- Studying nuclear DNA, mitochondrial DNA, ribosomal RNA, and proteins is particularly useful in evolutionary biology because these molecules can provide direct evidence of changes in genes and thus evolution.

Molecular Biology

- Just as animals can have homologous structures, animals also have homologous biochemical processes that can be studied using molecular biological methods.
- Molecular methods have several advantages:
- they are useful with all organisms, the data are quantifiable with readily available computer software, and databases of molecular information for many organisms are very large and growing.
- The use of molecular data allows biologists to investigate the causes of the genetic variation and molecular processes that influence evolution.

Molecular Biology

- These data also provide information for the construction of evolutionary trees (phylogenies).
- The principle behind molecular analysis is that closely related organisms will be genetically more similar than distantly related organisms.
- Genetic similarity or degree of relatedness is reflected in the variation (or lack of it) in the amino acids that comprise a protein or in the bases that comprise DNA.
- This genetic variation can be quantified in a number of ways.

Molecular Biology

- Genetic variation is often measured by the proportion of polymorphic loci in a population.
- A polymorphic locus is one where two or more alleles exist.
- For example, imagine a researcher examined 20 loci from representatives of two populations.
- In the first population, the researcher found that five of these loci had more than one allele.
- The proportion of polymorphic loci would be 5/20 or 0.25.

Molecular Biology

- In the second population, 10 of these loci had more than one allele, and the proportion of polymorphic loci was 0.5.
- Genetic variation could be greater in the second population for a number of reasons.
- For example, it could indicate greater time since divergence from an ancestor and thus more time for variations to accumulate, or it could indicate genetic mixing with more than one ancestral group.
- Documenting genetic variation is important in evolutionary studies because this variation is the fuel for natural selection.

Molecular Biology

- Techniques for isolating and manipulating DNA have provided very powerful tools for the analysis of genetic variation among groups of animals.
- The polymerase chain reaction (PCR) and automated DNA sequencers allow researchers to begin with very small amounts of sample DNA and quickly and inexpensively determine the base sequences of DNA and other genetic fingerprinting patterns.
- Variation in DNA in homologous genes and regions suggests relationships between genes and groups of organisms.

Molecular Biology

- The tree diagram on the inside of the front cover of this textbook is based largely on the study of variation in the base sequence of ribosomal RNA.

Interpreting the Evidence: Phylogeny and Common Descent

Phylogeny and Common Descent

- Scientists use the data gathered from studies just described to understand how organisms are related to each other.
- **Phylogeny** refers to the evolutionary relationships among species.
- It includes the depiction of ancestral species and the relationships of modern descendants of a common ancestor.
- These depictions involve the use of phylogenetic trees showing lines of descent.
- In the past, biologists relied mainly on the fossil record and anatomical traits in the construction of phylogenetic trees.

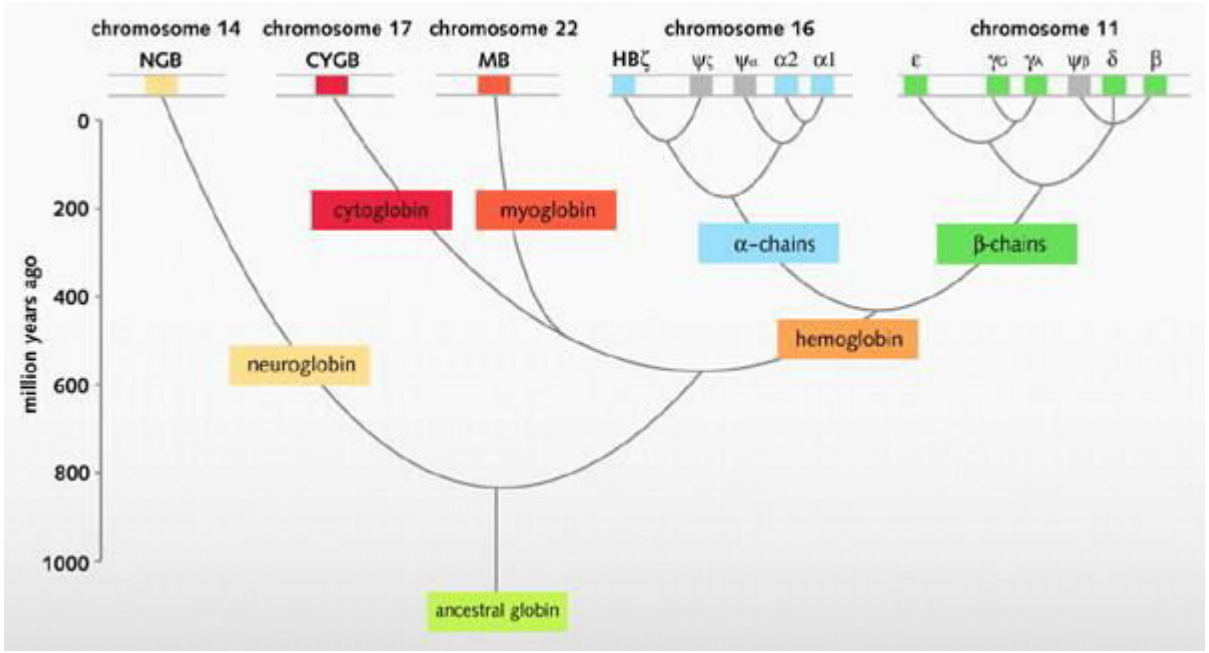
Phylogeny and Common Descent

- The addition of molecular data is revolutionizing phylogenetic studies.
- **Phylogenies** are reconstructed by examining variations in homologous structures, proteins, and genes.
- The process begins by gathering data from different organisms being compared.
- In molecular studies it usually involves sampling DNA or proteins from present-day organisms.
- Proteins or DNA from a particular locus or set of loci are compared using computer programs that include assumptions about the types of changes that are more likely to occur.

Phylogeny and Common Descent

- (Changes of purine to pyrimidine, or vice versa, are less likely than changes of one purine to another purine or one pyrimidine to another pyrimidine.)
- The computer program examines all possible relationships between the different organisms and groups the organisms based on the fewest number of evolutionary changes that must have taken place since they shared a common ancestor.
- Figure shows a phylogenetic tree for the hemoglobin molecule.
- Hemoglobin genes are grouped into alpha and beta families.

Phylogeny and Common Descent



Phylogeny and Common Descent

- The products of these genes are incorporated into the hemoglobin molecule that transports oxygen in red blood cells.
- Another related gene codes for the oxygen storage pigment in muscle, myoglobin.
- Genes in the beta family have branch points or nodes that represent ancestral molecules.
- In other trees, nodes could represent individuals, populations, or species.

Phylogeny and Common Descent

- A branch represents an evolutionary connection between molecules (individuals, populations, or species).
- The longer the branch, the greater the variation and the more distant the evolutionary relationship between molecules (individuals, populations, or species).
- The relatively short branch distance between nodes in the beta family reflects close evolutionary ties.
- Similar conclusions are drawn for genes in the alpha family.
- The conclusions drawn from this tree are that all of the modern hemoglobin genes are derived from a single gene that existed between 600 and 800 million years ago.

Phylogeny and Common Descent

- Phylogenetic trees like this one, and the many other, reinforce our ideas of common descent.
- All life is related, and the evidence of this relationship is overwhelming.
- Evolution is the major unifying theme in biology because it helps explain both the similarities and the diversity of life.
- There is no doubt that it has occurred in the past and continues to occur today.

Phylogeny and Common Descent

Modern Synthesis

- How the principles of population genetics have been combined with Darwinian evolutionary theory into what is often called the modern synthesis.
- The Modern Synthesis describes the **fusion** (merger) of Mendelian **genetics** with Darwinian evolution that resulted in a unified theory of evolution. It is sometimes referred to as the **Neo-Darwinian theory**. The Modern Synthesis was developed by a number of now-legendary evolutionary biologists in the 1930s and 1940s.

The modern synthesis: a closer look

the Hardy-Weinberg theorem

the Hardy-Weinberg theorem

Introduction

- In 1908, English mathematician Godfrey H. Hardy and German physician Wilhelm Weinberg independently derived a mathematical model describing what happens to the relative frequency of alleles in a sexually reproducing population over time.
- Their combined ideas became known as the **Hardy-Weinberg theorem.**

Statement

- It states that the mixing of alleles at meiosis and their subsequent recombination do not alter the relative frequencies of the alleles in future generations, if certain assumptions are met.

the Hardy-Weinberg theorem

- **Stated another way**, if certain assumptions are met, evolution will not occur because the relative allelic frequencies will not change from generation to generation, even though the specific mixes of alleles in individuals may vary.

the Hardy-Weinberg theorem

Assumptions

- The assumptions of the Hardy–Weinberg theorem are as follows:

1. **Population Size**

- The population size must be large. Large size ensures that gene frequency will not change by chance alone.

2. **Migration**

- Individuals cannot migrate into, or out of, the population. Migration may introduce new alleles into the gene pool or add or delete copies of existing alleles.

the Hardy-Weinberg theorem

3. **Mutational Equilibrium**

- Mutations must not occur.
- If they do, mutational equilibrium must exist.
- Mutational equilibrium exists when mutation from the wild-type allele to a mutant form is balanced by mutation from the mutant form back to the wild type.
- In either case, no new genes are introduced into the population from this source.

the Hardy-Weinberg theorem

4. Sexual Reproduction

- Sexual reproduction within the population must be random.
- Every individual must have an equal chance of mating with any other individual in the population.
- If this condition is not fulfilled, then some individuals are more likely to reproduce than others, and natural selection may occur.

the Hardy-Weinberg theorem

- These assumptions must be met if allelic frequencies are not changing—that is, if evolution is not occurring.
- Clearly, these assumptions are restrictive, and few, if any, real populations meet them.
- This means that most populations are evolving.
- The Hardy–Weinberg theorem, however, does provide a useful theoretical framework for examining changes in allelic frequencies in populations.
- The next section explains how, when the assumptions are not met, microevolutionary change occurs.

EVOLUTIONARY MECHANISMS

population size

EVOLUTIONARY MECHANISMS

- Evolution is neither a creative force working for progress nor a dark force sacrificing individuals for the sake of the group.
- It is neither moral nor immoral.
- It has neither a goal nor a mind to conceive a goal.
- Such goal-oriented thinking is said to be teleological.
- **Evolution** is simply a result of some individuals in a population surviving and being more effective at reproducing than others in the population, leading to changes in relative allelic frequencies.

EVOLUTIONARY MECHANISMS

- This section examines some of the situations when the Hardy–Weinberg assumptions are not met—situations in which gene frequencies change from one generation to the next and evolution occurs.

Population Size

Population

- A **population** is defined as a group of individuals of the same species living and interbreeding within a given area.
- Members of a population often rely on the same resources, are subject to similar environmental constraints, and depend on the availability of other members to persist over time.
- Scientists study a population by examining how individuals in that population interact with each other and how the population as a whole interacts with its environment.

Population Size

Demographic Parameters

- As a tool for **objectively** studying populations, population ecologists rely on a series of statistical measures, known as demographic **parameters**, to describe that population.

Demography

- The field of science interested in collecting and analyzing these numbers is termed population demographics, also known as **demography**.
- Broadly defined, **demography** is the study of the characteristics of populations.
- It provides a mathematical description of how those characteristics change over time.

Population Size

- **Demographics** can include any statistical factors that influence population growth or decline, but several parameters are particularly important:
- population size
- density
- age structure
- fecundity (birth rates)
- mortality (death rates)
- sex ratio

Population Size

- The most fundamental demographic **parameter** is the number of individuals within a population.
- **Population size** is defined as the number of individuals present in a **subjectively designated geographic range**.
- Despite the simplicity in its concept, locating all individuals during a census (a full count of every individual) is nearly impossible, so ecologists usually estimate population size by counting individuals within a small sample area and **extrapolating** that sample to the larger population.
- Regardless of the challenges in measuring population size, it is an important characteristic of a population with significant implications for the dynamics of the population as a whole.

Population Size

- Populations display distinctive behaviors based on their size. Small populations face a greater risk of **extinction**.
- Individuals in these populations can have a hard time finding quality mates so, fewer individuals mate and those that do risk **inbreeding**.
- Additionally, individuals in small population are more susceptible to random deaths.
- Events like fire, floods, and disease have a greater chance of killing all individuals in the population.
- Large populations experience their own problems.

Population Size

- As they approach the maximum **sustainable** population size, known as carrying capacity, large populations show characteristic behavior.
- Populations nearing their carrying capacity experience greater competition for resources, shifts in **predator-prey relationships**, and lowered fecundity.
- If the population grows too large, it may begin to exceed the carrying capacity of the environment and degrade available habitat.

EVOLUTIONARY MECHANISMS

Genetic Drift (Part I)

Genetic Drift

- Chance often plays an important role in the perpetuation of genes in a population, and the smaller the population, the more significant the chance may be.
- Fortuitous circumstances, such as a chance encounter between reproductive individuals, may promote reproduction.
- Some traits of a population survive not because they convey increased fitness, but because they happen to be in gametes involved in fertilization.

Genetic Drift

- Chance events influencing the frequencies of genes in populations result in **genetic drift**.

Genetic Drift

- The process of genetic drift is analogous to flipping a coin.
- The likelihood of getting a head or a tail is equal.
- The 50:50 ratio of heads and tails is most likely in a large number of tosses.
- In only 10 tosses, for example, the ratio may be a disproportionate 7 heads and 3 tails.
- Similarly, the chance of one or the other of two equally adaptive alleles being incorporated into a gamete, and eventually into an individual in a second generation, is equal.

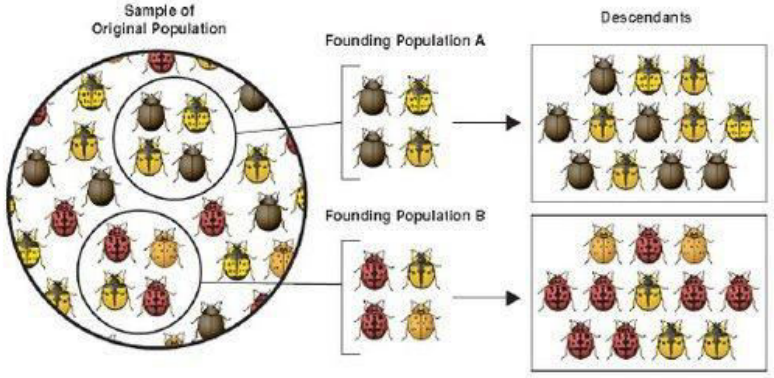
Genetic Drift

- Gamete sampling in a small population may show unusual proportions of alleles in any one generation of gametes because meiotic events, like tossing a coin, are random.
- Assuming that both alleles have equal fitness, these unusual proportions are reflected in the genotypes of the next generation.
- These chance events may result in a particular allele increasing or decreasing in frequency as shown in figure below.
- In small populations, inbreeding is also common. Genetic drift and inbreeding are likely to reduce genetic variation within a population.

Genetic Drift

Genetic Drift

- A random change in allele frequency



Genetic Drift In small populations, individuals that carry a particular allele may have more descendants than other individuals do. Over time, a series of chance occurrences of this type can cause an allele to become more common in a population. This model demonstrates how two small groups from a large, diverse population could produce new populations that differ from the original group.

Genetic Drift

- If a mutation introduces a new allele into a population and that allele is no more or less adaptive than existing alleles, genetic drift may permit the new allele to become established in the population, or the new allele may be lost because of genetic drift.
- When one member of a pair of alleles is lost, the surviving allele is said to be fixed in the population.
- Fixation of an allele can occur within just a few generations of random mating.

Genetic Drift

- The likelihood of genetic drift occurring in small populations suggests that a Hardy–Weinberg equilibrium will not occur and evolution is occurring.
- Two special cases promote genetic drift.
- **One** occurs when a subset of a larger population colonizes a new habitat and the **second** occurs when a population undergoes a near extinction event.

EVOLUTIONARY MECHANISMS

Genetic Drift (Part II)

The Founder Effect

- When a few individuals from a parental population colonize new habitats, they seldom carry alleles in the same frequency as the alleles in the gene pool from which they came.
- The new colony that emerges from the founding individuals is likely to have a distinctive genetic makeup with far less variation than the larger population.
- This form of genetic drift is the **founder effect**.
- An often-cited example of the founder effect concerns the genetic makeup of the Dunkers of eastern Pennsylvania.

The Founder Effect

- They emigrated from Germany to the United States early in the eighteenth century and, for religious reasons, have not married outside their sect.
- Examination of certain traits (e.g., ABO blood group) in their population reveals very different gene frequencies from the Dunker populations of Germany.
- These differences are attributed to the chance absence of certain genes in the individuals who founded the original Pennsylvania Dunker population.

The Bottleneck Effect

- A second special case of genetic drift can occur when the number of individuals in a population is drastically reduced.
- For example, cheetah populations in South and East Africa are endangered.
- Their depleted populations have reduced genetic diversity to the point that even if population sizes are restored, they will have only a remnant of the original gene pool.

The Bottleneck Effect

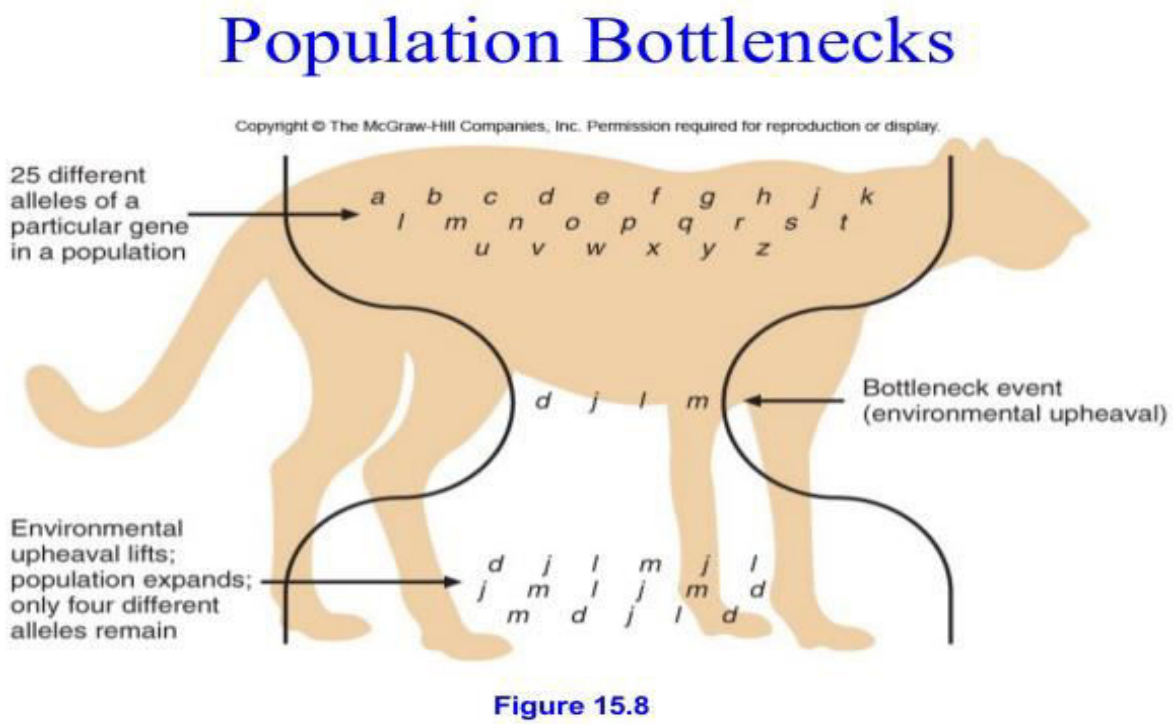


Figure 15.8

The Bottleneck Effect

- This form of genetic drift is called the **bottleneck effect**.
- A similar example concerns the northern elephant seal (*Mirounga angustirostris*), which was hunted to near extinction in the late 1800s for its blubber, which was used to make prized oil as shown in figure below.
- The population was reduced to about 100 individuals.
- Because males compete for reproductive rights, very few males actually passed their genes on to the next generation.

The Bottleneck Effect



The Bottleneck Effect

- Legislation to protect the elephant seal was enacted in 1922, and now the population is greater than 100,000 individuals.
- In spite of this relatively large number, the genetic variability in the population is very low.
- One study showed no genetic variation at 24 proteincoding loci.

The Bottleneck Effect

- The effects of bottlenecks are somewhat controversial.
- The traditional interpretation is that decreases in genetic diversity make populations less likely to withstand environmental stress and more susceptible to extinction.
- That is, a population with high genetic diversity is more likely to have some individuals with a combination of genes that allows them to withstand environmental changes.

The Bottleneck Effect

- In the case of the cheetah, recent evidence indicates that this cat's current problems may be more a result of lions and spotted hyenas preying on cheetah cubs than low genetic diversity.
- Most evolutionary biologists agree, however, that over evolutionary time frames, high genetic diversity makes extinction less likely.

EVOLUTIONARY MECHANISMS

Gene Flow

Gene Flow

- The Hardy–Weinberg theorem assumes that no individuals enter a population from the outside (immigrate) and that no individuals leave a population (emigrate).
- Immigration or emigration upsets the Hardy–Weinberg equilibrium, resulting in changes in relative allelic frequency (evolution).

Gene Flow

- Changes in relative allelic frequency from the migration of individuals are called **gene flow**.
- Although some natural populations do not have significant gene flow, most populations do.

Gene Flow

Effects of Gene Flow

- The effects of gene flow can differ, depending on the circumstances.
- The exchange of alleles between an island population and a neighboring continental population, for example, can change the genetic makeup of those populations.
- If gene flow continues and occurs in both directions, the two populations will become more similar.
- The absence of gene flow can make change in populations less likely. For example, evidence suggests that the African elephants should be divided into two species.

Gene Flow

- Elephants of Africa's tropical forests are smaller, have straighter and thinner tusks, and have distinctive skull morphology as compared to savannah elephants as shown in figure below.
- Molecular studies reveal marked genetic differences between these two groups.
- A lack of gene flow between these groups has helped maintain the unique allelic frequencies within the groups.

Gene Flow



Gene Flow

Gene Flow and African Elephants

- African elephants have been considered members of a single species, *Loxodonta africana*.
- Differences in morphology and habitats resulted in the description of two subspecies:
- (a) *L. africana* africana (savannah subspecies)
- (b) *L. africana cyclotis* (tropical forest subspecies).
- Evidence from molecular studies indicates that little gene flow exists between these two groups of elephants and suggests that they should be considered separate species: *L. africana* and *L. cyclotis*.

EVOLUTIONARY MECHANISMS

Mutation

Mutations

Definition

- **Mutations** are changes in the structure of genes and chromosomes.
- The Hardy–Weinberg theorem assumes that no mutations occur or that mutational equilibrium exists.
- Mutations, however, are a fact of life.
- Most important, mutations are the origin of all new alleles and a source of variation that may be adaptive for an animal.

Mutations

- Mutation counters the loss of genetic material from natural selection and genetic drift, and it increases the likelihood that variations will be present that allow some individuals to survive future environmental shocks.
- Mutations make extinction less likely.
- Mutations are random events, and the likelihood of a mutation is not affected by the mutation's usefulness.
- Organisms cannot filter harmful genetic changes from advantageous changes before they occur.

Mutations

- The effects of mutations vary enormously.

Neutral Mutations

- Neutral muta - tions are neither harmful nor helpful to the organism.
- Neutral mutations may occur in regions of DNA that do not code for proteins.
- Other neutral mutations may change a protein's structure, but some proteins tolerate minor changes in structure without affecting the function of the protein.
- Mutations that do affect protein function are more likely to be detrimental than beneficial.

Mutations

- This is true because a random change in an established protein upsets millions of years of natural selection that occurred during the protein's evolution.
- Mutations in DNA that is incorporated into a gamete have the potential to affect the function of every cell in an individual in the next generation.
- These mutations are likely to influence the evolution of a group of organisms.

Mutations

- **Mutational equilibrium** exists when a mutation from the wild-type allele to a mutant form is balanced by a mutation from the mutant back to the wild type.
- This has the same effect on allelic frequency as if no mutation had occurred.
- Mutational equilibrium rarely exists, however.
- **Mutation pressure** is a measure of the tendency for gene frequencies to change through mutation.

Natural Selection

Natural Selection

- The theory of natural selection remains preeminent in modern biology.
- Natural selection occurs whenever some phenotypes are more successful at leaving offspring than other phenotypes.
- The tendency for natural selection to occur— and upset Hardy–Weinberg equilibrium—is selection pressure.
- Although natural selection is simple in principle, it is diverse in operation.

Modes of Selection

- For certain traits, many populations have a range of phenotypes, characterized by a bell-shaped curve that shows that phenotypic extremes are less common than the intermediate phenotypes.
- Natural selection may affect a range of phenotypes in three ways.
- Directional selection occurs when individuals at one phenotypic extreme are at a disadvantage compared to all other individuals in the population.
- In response to this selection, the deleterious gene(s) decreases in frequency, and all other genes increase in frequency.
- Directional selection may occur when a mutation gives rise to a new gene, or when the environment changes to select against an existing phenotype.

Modes of Selection

- Industrial melanism, a classic example of directional selection, occurred in England during the Industrial Revolution.
- Museum records and experiments document how environmental changes affected selection against one phenotype of the peppered moth, Biston betularia.

Modes of Selection

- In the early 1800s, a gray form made up about 99% of the peppered moth population.
- That form still predominates in nonindustrial northern England and Scotland.
- In industrial areas of England, a black form replaced the gray form over a period of about 50 years.
- In these areas, the gray form made up only about 5% of the population, and 95% of the population was black.
- The gray phenotype, previously advantageous, had become deleterious.

Modes of Selection

- The nature of the selection pressure was understood when investigators discovered that birds prey more effectively on moths resting on a contrasting background.
- Prior to the Industrial Revolution, gray moths were favored because they blended with the bark of trees on which they rested.
- The black moth contrasted with the lighter, lichen-covered bark and was easily spotted by birds.
- Early in the Industrial Revolution, however, factories used soft coal, and spewed soot and other pollutants into the air.

Modes of Selection

- Soot covered the tree trunks and killed the lichens where the moths rested.
- Bird predators now could easily pick out gray moths against the black background of the tree trunk, while the black form was effectively camouflaged.
- In the 1950s, the British Parliament enacted air pollution standards that have reduced soot in the atmosphere.
- As expected, the gray form of the moth has experienced a small but significant increase in frequency.

Natural Selection

Modes of Selection

- Another form of natural selection involves circumstances selecting against individuals of an intermediate phenotype.
- Disruptive or diversifying selection produces distinct subpopulations.
- An interesting example of disruptive selection also illustrates another form of selection, sexual selection.
- Sexual selection occurs when individuals have varying success obtaining mates.

Modes of Selection

- It often results in the evolution of structures used in combat between males for mates, such as antlers and horns, or ornamentation that attracts individuals of the opposite sex, such as the brightly colored tail feathers of a peacock.
- Sexual selection is considered by some to be a form of natural selection, but others consider it separately from natural selection.

Modes of Selection

- The plainfin midshipman (Porichthys notatus) inhabit depths of up to 400 m along the Pacific coast of North America.
- They are named for rows of bioluminescent photophores that reminded some of the rows of buttons on a naval uniform.
- Males move to shallower water for reproduction.
- The male establishes a nest in rock crevices and cares for young after attracting a female and spawning.
- Males of the species have one of two body forms (morphs) designated as type I and type II males.

Modes of Selection

- Type I males have an exaggerated morphology that probably arose as a result of sexual selection.
- Their very large heads and wide mouths are used in defending territories and nests under rocks.
- Their exaggerated sonic musculature is used to court females with loud humming vocalizations that can be heard by boaters and beachcombers.
- Aggressive vocalizations are used to ward off other males.

Modes of Selection

- The larger the males, the more effective they are in establishing large nests and providing long-term care for young.
- Type II males have a smaller, drab morphology and look similar to females.
- These males are sometimes called “sneakers” because they dart into type I males’ nests and fertilize eggs without having to invest their time and energy into nest construction, defense, and mate attraction.
- The type II morphology is apparently not recognized readily by type I males as a threat to their reproductive success.
- Type II males are tolerated near the nests of type I males, which improves the chances that type II males succeed with their “sneaker” strategy.

Modes of Selection

- Disruptive selection favors the retention of the reduced type II morphology because of the energy savings in not having to establish and defend a nest and in not supporting a larger body size.
- Disruptive selection also favors the retention of the exaggerated type I morphology because that is the body form that can establish territories and nests, defend them from other type I males, and attract females.
- When both phenotypic extremes are deleterious, a third form of natural selection—stabilizing selection— narrows the phenotypic range.

Modes of Selection

- During long periods of environmental constancy, new variations that arise, or new combinations of genes that occur, are unlikely to result in more fit phenotypes than the genes that have allowed a population to survive for thousands of years, especially when the new variations are at the extremes of the phenotypic range.
- A good example of stabilizing selection is the horseshoe crab (Limulus), which lives along the Atlantic coast of the United States.
- Comparison of the fossil record with living forms indicates that this body form has changed little over 200 million years.

Modes of Selection

- Apparently, the combination of characteristics present in this group of animals is adaptive for the horseshoe crab's environment.

DE VRIES MUTATION THEORY

What is a Mutation?

- Mutation is a **change in the nucleotide sequence** of **a gene or a chromosome.**
- In biology, a mutation is **the permanent alteration** of **the nucleotide sequence of the genome** of an organism, virus, or extra-chromosomal DNA or other genetic elements.
- Mutations result from **errors during DNA replication** or **other types of damage to DNA** (such as may be caused by exposure to **radiation** or **carcinogens**).

Some history. . . .

- Hugo de Vries, Principles of the theory of mutation. Science (1914)
  - First coined the term "mutation"
  - Idea that genetic variation arises from some unspecified "intracellular perturbation"
- Hermann J. Muller, Artificial transmutation of the gene, Science (1927)
  - X-ray mutagenesis of Drosophila
  - In a single experiment increased the known number of Drosophila mutants by 50%

Hugo De Veirs

In 1901, **Hugo DeVries**, suggested that inherited mutations caused variations.

He believed that mutations (changes in genetic material) occurred randomly and those mutations that were favorable were inherited by offspring.

DeVries based his theory on his reproduction experiments with the evening primrose plant.



HUGO DeVRIES: Theory of Mutation (1901)

evening primrose plants





The theory of mutation of Hugo De Vries.

1. Mutations or discontinuous variations are **the raw material of evolution**.
2. Mutations **appear all of a sudden**. They become **operational immediately**.
3. Unlike Darwin's continuous variations or fluctuations, mutations do not **revolve around the mean or normal character of the species.**
4. The same type of mutations **can appear in a number of individuals of a species.**
5. **All mutations are inheritable.**
6. Mutations **appear in all conceivable directions**.
7. Useful mutations are **selected by nature**. Lethal mutations are **eliminated**. However, **useless and less harmful ones can persist in the progeny.**
8. Accumulation of variations **produce new species**. Sometimes a new species is produced from a **single mutation**.
9. **Evolution is a jerky and discontinuous process.**

Points in Favor of the Mutation Theory

1. Mutations are **actually the source** of **all variations and hence fountain head of evolution.**
2. Mutation theory can explain **both progressive and retrogressive evolution.**
3. As the ratio of mutations is not the same in all individuals and their parts, mutation theory can **explain the occurrence of both changed and unchanged forms.**
4. A number of mutations have appeared in the past.

Criticism of the Mutation Theory

1. Natural mutations are not common as Hugo de Vries thought them to occur.
2. Most of the mutations are negative or retrogressive.
3. Mutations are generally recessive while traits taking part in evolution are usually dominant.
4. Mutation theory cannot satisfactorily explain the development of mimicry, mutual dependence of flowers and pollinating insects.
5. This theory does not explain the role of nature.

RATES OF EVOLUTION

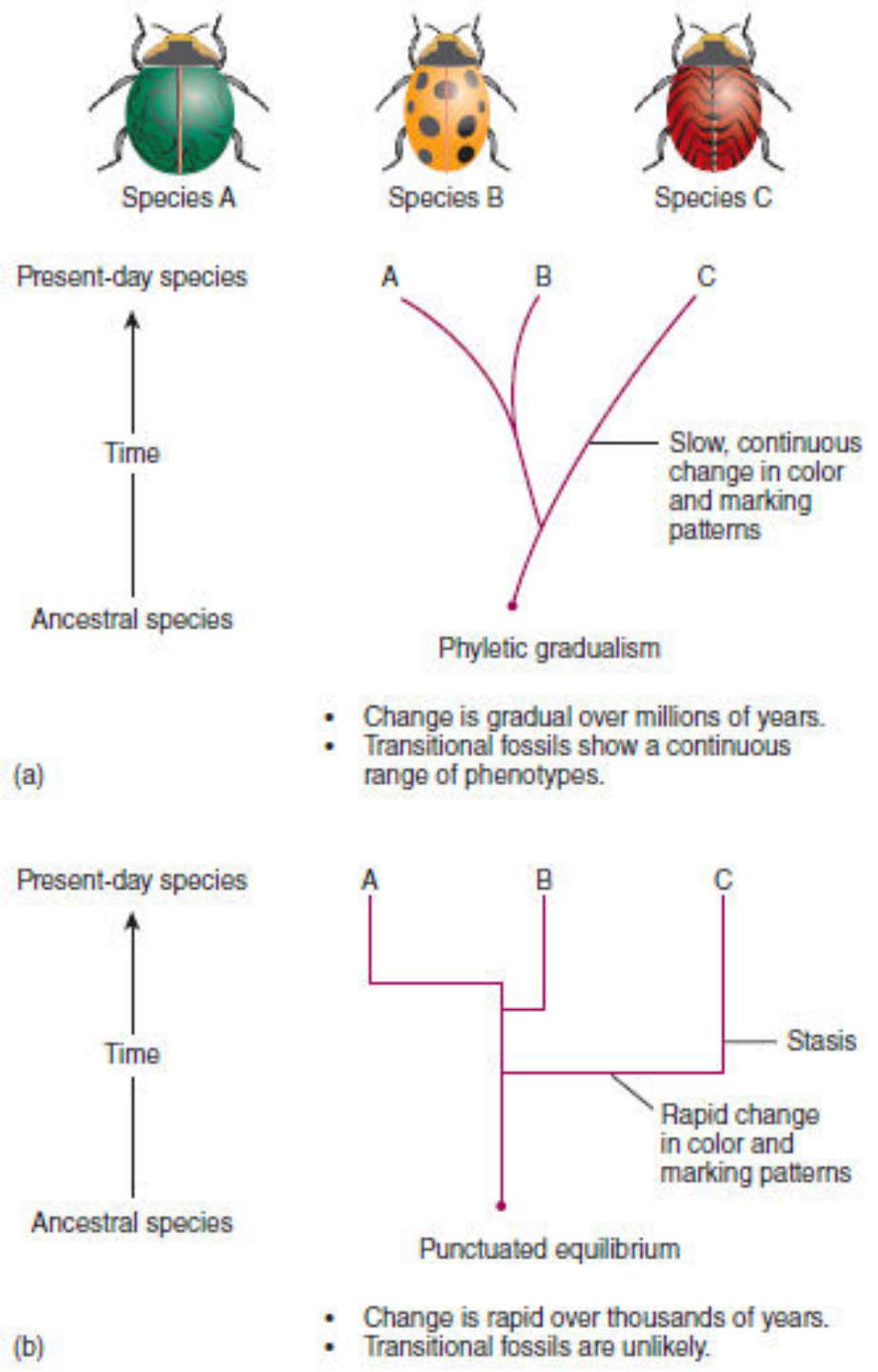
RATES OF EVOLUTION

Rates of Evolution

- Charles Darwin perceived evolutionary change as occurring gradually
- over millions of years. This concept, called **phyletic gradualism**, has been the traditional interpretation of the tempo, or rate, of evolution.
- Some evolutionary changes, however, happen very rapidly.
- Studies of the fossil record show that many species do not change significantly over millions of years.
- These periods of stasis (Gr. stasis, standing still), or equilibrium, are interrupted when a group encounters an ecological crisis, such as a change in climate or a major geological event.
- Over the next 10,000 to 100,000 years, a variation that previously was selectively neutral or disadvantageous might now be advantageous.

Rates of Evolution

- Alternatively, geological events might result in new habitats becoming available. (Events that occur in 10,000 to 100,000 years are almost instantaneous in an evolutionary time frame.) This geologically brief period of change "punctuates" the previous million or so years of equilibrium and eventually settles into the next period of stasis (figure).
- Long periods of stasis interrupted by brief periods of change characterize the **punctuated equilibrium model** of evolution.



Rates of Evolution. A comparison of phyletic gradualism and punctuated equilibrium in three hypothetical beetle species. (a) In the phyletic gradualism model of evolution, changes are gradual over long time periods. Note that this tree implies a gradual change in color and marking patterns in the three beetle species. (b) In the punctuated equilibrium model of evolution, rapid periods of change interrupt long periods of stasis. This tree implies that the color and marking patterns in the beetles changed rapidly and did not change significantly during long periods of stabilizing selection (stasis).

Rates of Evolution

- Biologists have observed such rapid evolutionary changes in small populations.
- In a series of studies over a 20-year period, Peter R. Grant has shown that natural selection results in rapid morphological changes in the bills of Galápagos finches.
- A long, dry period from the middle of 1976 to early January 1978 resulted in birds with larger, deeper bills.
- Early in this dry period, birds quickly consumed smaller, easily cracked seeds. As they were forced to turn to larger seeds, birds with weaker bills were selected against, resulting in a measurable change in the makeup of the finch population of the island Daphne Major.

Rates of Evolution

- The evolution of multiple species of cichlid fish in Lake Victoria (in the last 14,000 years is another example of rapid evolutionary change leading to speciation.
- One cichlid species (*Haplochromis pyrrhocephalus*) was nearly extirpated from Lake Victoria following the introduction of the Nile perch.
- Subsequent fishing pressure on the Nile perch allowed this cichlid population to recover somewhat; however, decreased water quality and oxygen levels placed additional selection pressures on its survival.
- Over a brief span of 20 years, researchers have documented a 64% increase in gill surface area and corresponding changes in head morphology that are allowing this cichlid population to recover once again.

Rates of Evolution

- Periods of stasis in the punctuated equilibrium model may be the result of stabilizing selection during times when environmental conditions are not changing.
- The ability of some organisms to escape changing environments through migration when environments are changing may also promote stasis.

Rates of Evolution

- One advantage of the **punctuated equilibrium model** is its explanation for the fossil record not always showing transitional stages between related organisms.
- The absence of transitional forms can often be attributed to fossilization being an unlikely event; thus, many transitional forms disappeared without leaving a fossil record.
- Because punctuated equilibrium involves rapid changes in small, isolated populations, preservation of intermediate forms in the fossil record is even less likely.
- The rapid pace (geologically speaking) of evolution resulted in apparent "jumps" from one form to another.

Rates of Evolution

- Phyletic gradualism and punctuated equilibrium are both valid models that explain evolutionary rates. Gradualism best describes the evolutionary history of some groups (e.g., mammals).
- Punctuated equilibrium best describes the evolutionary history of others (e.g., some marine invertebrates).
- For still other groups, evolution has been accented by periods when gradualism prevailed and other periods of rapid change and stasis.

POLYMORPHISM

Polymorphism

Polymorphism

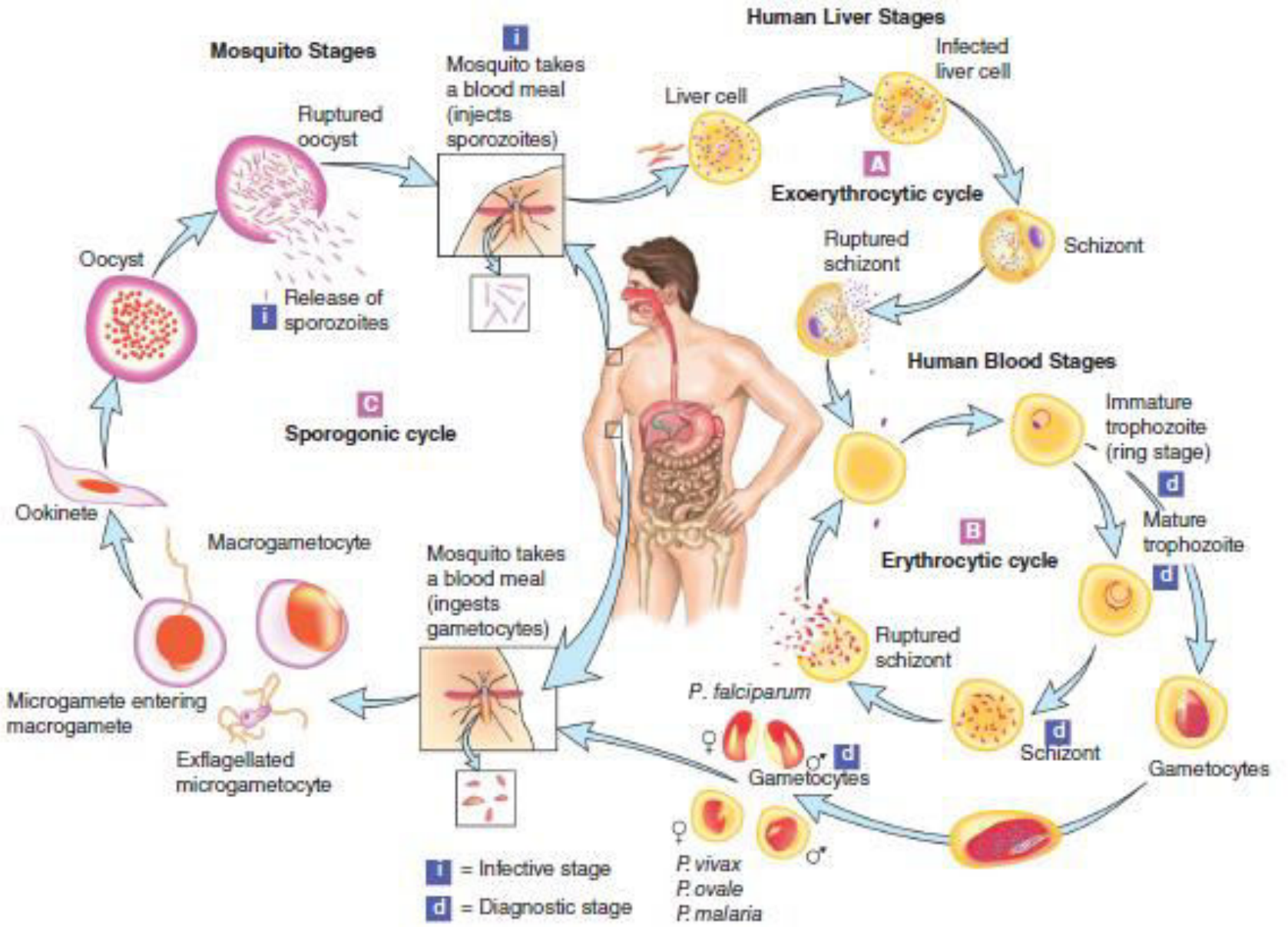
- **Polymorphism** occurs in a population when two or more distinct forms exist without a range of phenotypes between them.

Balanced polymorphism

- **Balanced polymorphism** (Gr. poly, many 1 morphe, form) occurs when different phenotypes are maintained at relatively stable frequencies in the population and may resemble a population in which disruptive selection operates.
- Sickle-cell anemia results from a change in the structure of the hemoglobin molecule. Some of the red blood cells of persons with the disease are misshapen, reducing their ability to carry oxygen.
- In the heterozygous state, the quantities of normal and sickled cells are roughly equal.
- Sickle-cell heterozygotes occur in some African populations with a frequency as high as 0.4.
- The maintenance of the sickle-cell heterozygotes and both homozygous genotypes at relatively unchanging frequencies makes this trait an example of a balanced polymorphism.

Balanced polymorphism

- Why hasn't natural selection eliminated such a seemingly deleterious allele? The sickle-cell allele is most common in regions of Africa that are heavily infected with the malarial parasite *Plasmodium falciparum*.
- This parasite is transmitted by mosquitoes and has a life cycle that involves the invasion of red blood cells and liver cells (**see figure**).
- Symptoms of the disease include recurring bouts of chills and fever, and it remains one of the greatest killers of humanity.
- Sickle-cell heterozygotes are less susceptible to malarial infections; if infected, they experience less severe symptoms than do homozygotes without sickled cells.
- Individuals homozygous for the normal allele are at a disadvantage because they experience more severe malarial infections, and individuals homozygous for the sickle-cell allele are at a disadvantage because they suffer from the severe anemia that the sickle cells cause.



The Life Cycle of Plasmodium. Schizogony (merogony) occurs in liver cells and, later, in the red blood cells (RBCs) of humans. Gametogony occurs in RBCs. During a blood meal, the mosquito takes in micro- and macrogametes, which fuse to form zygotes. Zygotes penetrate the gut of the mosquito and form oocysts. Meiosis and sporogony form many haploid sporozoites that may enter a new host when the mosquito bites the host.

Balanced polymorphism

- The heterozygotes, who usually experience no symptoms of anemia, are more likely to survive than either homozygote.
- This system is also an example of heterozygote superiority—when the heterozygote is more fit than either homozygote.
- Heterozygote superiority can lead to balanced polymorphism because perpetuation of the alleles in the heterozygous condition maintains both alleles at a higher frequency than would be expected if natural selection acted only on the homozygous phenotypes.

Balanced polymorphism

- The processes that change relative allelic frequencies in populations can, over geological time periods (measured in thousands of years), result in the formation of new species.
- Species can become extinct very quickly when climatic or geological events cause abrupt environmental changes.
- The formation of new species and the extinction of species were defined in chapter 4 as macroevolution.
- Even though the formation of new species is difficult to observe directly because of the long time frames involved, the evidence has convinced the vast majority of scientists that macroevolution occurs.
- The rest of this chapter continues the theme of macroevolutionary change.

SPECIES AND SPECIATION

Species and speciation

Species and speciation

- Taxonomists classify organisms according to their similarities and differences.
- The fundamental unit of classification is the species.
- Unfortunately, formulating a universally applicable definition of species is difficult.
- According to the biological species concept, a **species** is a group of populations in which genes are actually, or potentially, exchanged through interbreeding.
- Although concise, this definition has problems associated with it. Taxonomists often work with morphological characteristics, and the reproductive criterion must be assumed based on morphological and ecological information.
- Also, some organisms do not reproduce sexually. Obviously, other criteria need to be applied in these cases.

Species and speciation

- Another problem concerns fossil material. Paleontologists describe species of extinct organisms, but how can they test the reproductive criterion?
- Finally, populations of similar organisms may be so isolated from each other that gene exchange is geographically impossible.
- To test a reproductive criterion, biologists can transplant individuals to see if mating can occur, but mating of transplanted individuals does not necessarily prove what would happen if animals were together in a natural setting.
- An increasingly popular phylogenetic species concept depicts a species as a group of populations that have evolved independently of other groups of populations. Species groups are studied using phylogenetic analysis principles.
- Rather than trying to establish a definition of a species that solves all these problems, it is probably better to simply understand the problems associated with the biological definition.
- In describing species, taxonomists use morphological, physiological, embryological, behavioral, molecular, and ecological criteria, realizing that all of these have a genetic basis.

Species and speciation

- **Speciation** is the formation of new species.
- A requirement of speciation is that subpopulations are prevented from interbreeding.
- For some reason, gene flow between populations or subpopulations does not occur. This is called **reproductive isolation**.
- When populations are reproductively isolated, natural selection and genetic drift can result in evolution taking a different course in each subpopulation.
- Reproductive isolation can occur in different ways.

Species and speciation

- Some forms of isolation may prevent mating from occurring. For example, barriers such as rivers or mountain ranges may separate subpopulations.
- Other forms of isolation may be behavioral. If the courtship behavior patterns of two animals are not mutually appropriate, or mating periods differ, mating does not occur.
- Other forms of isolation prevent successful fertilization and development even though mating
- may have occurred. Conditions within a female's reproductive tract may prevent fertilization by sperm of another species.
- The failure of hybrids to produce offspring of their own is a form of isolation that occurs even though mating and fertilization do occur.
- This hybrid nonviability is caused by structural differences between chromosomes—preventing the formation of viable gametes because chromosomes cannot synapse properly during meiosis.

Allopatric Speciation

- **Allopatric** (Gr. allos, other 1 patria, fatherland) **speciation** occurs when subpopulations become geographically isolated from one another.
- For example, a mountain range or river may permanently separate members of a population.
- Adaptations to different environments or genetic drift in these separate populations may result in members not being able to mate successfully with each other, even if experimentally reunited.
- Many biologists believe that allopatric speciation is the most common kind of speciation.

Parapatric Speciation

- Another form of speciation, called **parapatric** (Gr. para, beside) **speciation**, occurs in small, local populations, called **demes**.
- For example, all of the frogs in a particular pond or all of the sea urchins in a particular tidepool make up a deme.
- Individuals of a deme are more likely to breed with one another than with other individuals in the larger population, and because they experience the same environment, they are subject to similar selection pressures.
- Demes are not completely isolated from each other because individuals, developmental stages, or gametes can move among demes of a population.
- On the other hand, the relative isolation of a deme may mean that its members experience different selection pressures than other members of the population. If so, speciation can occur.

Parapatric Speciation

- Although most evolutionists theoretically agree that parapatric speciation is possible, no certain cases are known.
- Parapatric speciation is therefore considered of less importance in the evolution of animal groups than allopatric speciation.

Sympatric Speciation

- A third kind of speciation, called **sympatric** (Gr. sym, together) **speciation**, occurs within a single population.
- Even though organisms are sympatric, they still may be reproductively isolated from one another.
- In order to demonstrate sympatric speciation, researchers must demonstrate that two species share a common ancestor and then that the two species arose without any form of geographic isolation.
- The latter is especially difficult to demonstrate.
- The driving forces for speciation are difficult to reconstruct because current ecological and selective factors may not reflect those present in the evolutionary past.
- In spite of these difficulties, evidence is mounting that sympatric speciation plays a larger role in speciation than previously thought.

MOLECULAR EVOLUTION

MOLECULAR EVOLUTION

Molecular Evolution

- Many evolutionists study changes in animal structure and function that are observable on a large scale—for example, changes in the shape of a bird's bill or in the length of an animal's neck.
- All evolutionary change, however, results from changes in the base sequences in DNA and amino acids in proteins.
- Molecular evolutionists investigate evolutionary relationships among organisms by studying DNA and proteins. For example, cytochrome c is a protein present in the cellular respiration pathways in all eukaryotic organisms.

Molecular Evolution

- Cellular respiration is the set of metabolic pathways that convert energy in organic molecules, such as the simple sugar glucose, into energy tied up in the bonds of adenosine triphosphate (ATP).
- ATP is the form of chemical energy immediately useful in cells.
- Organisms that other research has shown to be closely related have similar cytochrome c molecules. That cytochrome c has changed so little during hundreds of millions of years does not suggest that mutations of the cytochrome c gene do not occur.
- Rather, it suggests that mutations of the cytochrome c gene are nearly always detrimental and are selected against.
- Because it has changed so little, cytochrome c is said to have been conserved evolutionarily and is very useful for establishing relationships among distantly related organisms.

Molecular Evolution

- Not all proteins are conserved as rigorously as cytochrome c.
- Some regions of DNA that do not code for proteins can change without detrimental effects and accumulate base changes over relatively short periods of time.
- Comparing these regions of DNA can provide information on the relationships among closely related organisms.

Gene Duplication

- Recall that most mutations are selected against. Sometimes, however, an extra copy of a gene is present. One copy may be modified, but as long as the second copy furnishes the essential protein, the organism is likely to survive.
- **Gene duplication**, the accidental duplication of a gene on a chromosome, is one way that extra genetic material can arise.

MOSAIC EVOLUTION

MOSAIC EVOLUTION

Mosaic Evolution

- Rates of evolution can vary both in populations and in molecules and structures.
- A species is a mosaic of different molecules and structures that have evolved at different rates.
- Some molecules or structures are conserved in evolution; others change more rapidly.
- The basic design of a bird provides a simple example.
- All birds are easily recognizable as birds because of highly conserved structures, such as feathers, bills, and a certain body form.
- Particular parts of birds, however, are less conservative and have a higher rate of change. Wings have been modified for hovering, soaring, and swimming. Similarly, legs have been modified for wading, swimming, and perching.
- These are examples of mosaic evolution.

Final Term Syllabus
Lecture 36 to 69

Efforts By

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